



Measures of Location and Spread

• Mark Scheme

Statistics Chapter 2: Measures of Location and Spread. M = method mark A = accuracy mark (depends on M) B = independent mark ft = follow through

Section A – Central tendency

- 1 Mean = $65 \div 6 = 10.8$ B1; median = $\frac{9+12}{2} = 10.5$ B1.
- 2 $\sum fx = 83$, mean = 2.075 M1 A1; median (20th/21st values) = 2, mode = 2 B1.
- 3 Midpoints 5, 15, 25, 35, 45 B1; $\sum fx = 940$ M1; estimated mean = 23.5 kg A1.
- 4 (a) $\frac{n}{2} = 20$, median class $20 \leq m < 30$ B1; $20 + \frac{20-15}{15} \times 10$ M1; = 23.3 kg A1. (b) The exact values are unknown; midpoints/interpolation assume values are spread evenly in each class B1.
- 5(a) $11 \times 13 - 10 \times 12$ M1; $x = 23$ A1.
- 5(b) Midpoints 2.5, 7.5, 15, 30; $\sum fx = 480$ M1; estimated mean 12 min A1; $\frac{n}{2} = 20$, cumulative frequency reaches 20 at the end of $5 \leq t < 10$ M1; estimated median 10 min A1.
- 5(c) The data are skewed by a few long waiting times, which increase the mean B1.

Section B – Spread

- 6 Range = 17 B1; $Q_1 = 5.5$, $Q_3 = 12.5$, IQR = 7 B1.
- 7 Mean = 12 B1; $\sigma^2 = 158 - 144 = 14$ M1; $\sigma = 3.74$ A1.
- 8 $\sum fx = 91$, $\sum fx^2 = 317$ B1; mean = 3.03 B1; $\sigma = \sqrt{\frac{317}{30} - 3.03^2} = 1.17$ A1.
- 9 $\sum fx^2 = 26600$ M1 A1; $\sigma^2 = 665 - 23.5^2 = 112.75$ M1; $\sigma = 10.6$ kg A1.
- 10(a) A: $\sum x = 240$, $\sum x^2 = 12(3^2 + 20^2) = 4908$ M1 A1; B: $\sum x = 200$, $\sum x^2 = 8(4^2 + 25^2) = 5128$ A1.
- 10(b) Mean = $\frac{440}{20} = 22$ B1; $\sigma^2 = \frac{10036}{20} - 22^2$ M1; = 17.8 A1; $\sigma = 4.22$ A1.

Section C – Coding

- 11 Mean 53.2 B1; sd 1.5 B1.
- 12 $\bar{x} = 200 + 10(4.5) = 245$ M1 A1; $\sigma_x = 21$ A1.
- 13 $\bar{y} = 1.8$ B1; $\sigma_y = \sqrt{5.5 - 3.24} = 1.503$ M1; $\bar{x} = 1090$ g A1; $\sigma_x = 75.2$ g A1.
- 14 Mean = $1.8(15) + 32 = 59^\circ\text{F}$ M1 A1; sd = $1.8 \times 4 = 7.2^\circ\text{F}$ A1.
- 15(a) $\bar{y} = 13$ B1; $\sigma_y^2 = 300 - 169 = 131$ M1; $\sigma_y = 11.4$ A1.
- 15(b) Mean = $150000 + 1000(13) = £163000$ M1 A1; sd = £11400 A1.
- 15(c) New $y = 100$, so $\sum y = 620$, $n = 41$ M1; mean = $150000 + 1000 \times \frac{620}{41} = £165122$ A1.

Section D – Outliers and skewness

- 16 IQR = 12; upper limit $32 + 18 = 50$ M1; $55 > 50$, so 55 is an outlier A1.
- 17 (a) Mean > median > mode, so positive skew B1 B1. (b) The median, since it is not affected by the skew B1.
- 18 $Q_1 = 5$, median 6, $Q_3 = 8$ B1 B1; IQR = 3, upper limit $8 + 4.5 = 12.5$ M1; $25 > 12.5$, so an outlier A1.
- 19 Limits 50 ± 12 : 38 and 62 M1 A1; both 35 and 63 are outliers A1.
- 20(a) IQR = 0.13; upper limit $0.45 + 0.195 = 0.645$ M1; $0.82 > 0.645$ A1.



20(b) $Q_3 - Q_2 = 0.09 > Q_2 - Q_1 = 0.04$ *M1*; positive skew *A1*.

20(c) $0.39 + 2(0.12) = 0.63 < 0.82$, so yes *M1 A1*; median and IQR, as they are not affected by the outlier or the skew *B1*.

Roar-level challenge

21 (a) Mean = $20 + 1.5 = 21.5$ *B1*; $\sigma^2 = 25 - 1.5^2 = 22.75$, $\sigma = 4.77$ *M1 A1*. (b) $\sum(x - 20) = 25$, $\sum(x - 20)^2 = 350$ *M1*; mean = 22.3, $\sigma = \sqrt{\frac{350}{11} - \left(\frac{25}{11}\right)^2} = 5.16$ *A1*.